Highlights

Model-Free PID Tuning via Encirclements of the Step Response

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- A deterministic reading of the step response replaces manual PID tuning.
- Three phase portraits of the control error give a dimensionless damping diagnostic.
- The diagnostic is band-selective, assigning oscillation to the I, P, or D channel.
- A well-posedness guard keeps the encirclement count independent of window length.
- One triangular rule converges to the boundary of the well-damped tuning set.

Model-Free PID Tuning via Encirclements of the Step Response

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Abstract

Industrial PID tuning is still, at bottom, a visual craft: step the setpoint, look at the response, nudge a gain. This paper makes the looking exact. From one closed-loop step response, three phase portraits of the control error are formed—error against its running integral, first difference, and second difference—and the winding number of each about its settling point is counted. The three counts are proven invariant to time and amplitude scaling, so fixed dimensionless thresholds serve every loop, and they are shown to be band-selective: they detect insufficient damping in the low-, mid-, and high-frequency regions of the loop, the regions dominated by the integral, proportional, and derivative channels respectively. A well-posedness condition is established—without a settling-anchored observation window the counts diverge with window length—together with a one-parameter guard that restores window independence. A single triangular rule then updates the gains, ordered by the frequency band each channel dominates: raise every clean band below the lowest violation, cut that band—the only violation whose blame is unambiguous—and touch nothing above; or raise all bands when every portrait is clean. The iteration is deterministic, bounded by construction, and, under an explicit monotonicity assumption, converges to a limit cycle on the boundary of the well-damped tuning set. The method rests on two standing assumptions only—a stable, self-regulating process of the first-order-plus-time-delay class, and the parallel PID structure itself—and uses no process model, no solver, and one step test per iteration. It is validated on a battery of process-typical plants—lag-dominant, delay-dominant, balanced, and high-order—with identical defaults throughout. A reference implementation is public.

Keywords: PID control, controller tuning, data-driven control, phase plane, step response

1. Introduction

Eighty years after Ziegler and Nichols [1], PID remains the dominant industrial controller [2], and most PID loops are still tuned by inspection: an engineer steps the setpoint, looks at the response, and adjusts a gain by feel. The literature's two remedies both discard the inspection. Model-based rules—Ziegler–Nichols [1], Cohen–Coon [3], IMC [4], SIMC [6]—replace it with an identified low-order model and inherit that model's errors. Data-driven schemes replace it with a criterion and the machinery to serve it: a dedicated relay experiment that deliberately excites a sustained oscillation [5], gradient experiments on a quadratic cost [7], a one-shot identification-like problem [8], or online perturbation of the gains [9]. The inspection itself—which gain is to blame, given this response—has remained tacit.

This paper formalizes the inspection. The contributions are: (i) a triple of phase portraits of the control error whose encirclement counts form a dimensionless, scale-free, band-selective damping diagnostic (Section 2); (ii) a well-posedness analysis of the counts, including a counterexample showing that a naïvely windowed count diverges, and a settling-anchored guard that provably restores window independence (Section 3); (iii) a single triangular tuning rule in band-ordered gain coordinates, interpreted as band-wise loop shaping, with boundedness unconditional and boundary convergence under one explicit monotonicity assumption (Section 4); (iv) validation on a battery of process-typical plants with identical default parameters throughout (Section 5).

The method claims no optimality: it returns a fast, well-damped tuning—the contract an expert tuner offers—in exchange for one closed-loop setpoint step per iteration and nothing else. No model, no relay experiment, no solver; and full auditability, since an op-

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erator watching the three plots sees exactly what the algorithm sees. Figure 1 shows the entire method on one plant, before any mathematics.

2. The encirclement features

The method rests on exactly two standing assumptions, stated once and used throughout.

Assumption 1 (Process class)**.** *The process is a stable, self-regulating SISO plant whose open-loop step response is essentially monotone with dead time—the class adequately captured by first-order-plus-time-delay (FOPTD) models. FOPTD is meant here as a modelling class, not a literal transfer-function restriction: higher-order overdamped dynamics belong to it.*

Assumption 2 (Controller structure)**.** *The controller is a parallel PID, $u = K_p e + K_i \int e + K_d \dot{e}$, in a unity-feedback loop.*

Nothing else is assumed. No model is identified, parametrized, or used at any point of the procedure; no noise model is posited; no excitation is required beyond routine setpoint steps under the existing controller.

A tuning experiment applies a setpoint step and records the error e_k, $k = 0, \ldots, K$, at period T_s, over a window in which the response settles. Form the running integral E_k and the differences Δe_k, $\Delta^2 e_k$, and pair each signal with its own difference:

$$\begin{aligned} \Gamma_0 &: (e_k,\ E_k - E_K), \\ \Gamma_1 &: (\Delta e_k,\ e_k), \\ \Gamma_2 &: (\Delta^2 e_k,\ \Delta e_k). \end{aligned} \tag{1}$$

Γ_1 is the classical phase plane of the error; Γ_0 and Γ_2 are its integrated and differentiated siblings; all three terminate at the origin when the loop settles. We call (1) *Pachner plots*, after the first author, who devised the construction; the name has been in use in the authors' group since the construction was first implemented.

A well-damped loop spirals into the origin without completing a turn; an under-damped loop encircles it. The count is made precise as follows.

Definition 1 (Encirclement count)**.** *Given a planar trajectory (x_k, y_k), $k = 0, \ldots, J$: (i) normalize each coordinate by its peak magnitude, mapping the curve into $[-1, 1]^2$; (ii) truncate at the last entry into the disc of radius $\varepsilon \in (0, 1)$ about the origin; (iii) define N as the winding number of the remaining curve about the origin: total swept angle over 2π, evaluated as the endpoint angle difference corrected by signed crossings of the negative horizontal semi-axis. Partial revolutions count fractionally.*

Applying Definition 1 to $\Gamma_0, \Gamma_1, \Gamma_2$ yields the feature vector $\mathbf{N}(\theta) = (N_0, N_1, N_2)$ at controller setting $\theta = (K_p, K_i, K_d)$, from a single step test, with no plant knowledge.

Proposition 1 (Invariance)**.** *$\mathbf{N}$ is invariant under time reparametrization $t \mapsto \alpha t$, $\alpha > 0$, of the underlying response, and under independent positive scaling of either coordinate of each trajectory—in particular under scaling of the setpoint step and, in the fast-sampling limit, of the sampling period.*

Proof. The winding number depends only on the oriented image of the curve, not its parametrization; per-axis peak normalization removes amplitude and units, and the truncation disc is defined on normalized coordinates. □

Consequently one threshold set serves all loops: the defaults $\bar{\mathbf{N}} = (0.5, 0.75, 1.0)$ together with $\varepsilon = 0.1$ are used unchanged in every result of this paper. They function as the definition of "well damped," not as tuning knobs.

Proposition 2 (Band selectivity)**.** *Let the error contain a decaying mode $Ae^{-\sigma t} \cos \omega t$, $\sigma \ll \omega$. Differentiation weights the mode's amplitude by ω and integration by $1/\omega$, so the mode's visibility relative to the monotone component of the error is greatest in Γ_2 for large ω, in Γ_0 for small ω, and in Γ_1 near the closed-loop crossover. In the portrait it dominates, the mode traces a logarithmic spiral executing $N \approx \frac{\omega}{2\pi\sigma} \ln \frac{1}{\varepsilon'}$ revolutions before entering the truncation disc, where ε' is the disc radius relative to the mode's initial normalized amplitude: the count is a monotone image of the mode's inverse damping ratio, gated by frequency band.*

Inverting the small-damping estimate of Proposition 2 ($\zeta \approx \sigma/\omega$, so $\omega/\sigma \approx 1/\zeta$) gives an illustrative reading of the defaults as damping specifications: $\zeta \approx \ln(1/\varepsilon)/(2\pi \bar{N}_k)$, or roughly $\zeta \approx 0.73, 0.49, 0.37$ for $\bar{N}_0, \bar{N}_1, \bar{N}_2$ at $\varepsilon = 0.1$—decreasing across bands, consistent with the lowest band being the least tolerant of oscillation. The figures are indicative rather than exact, since the underlying approximation assumes $\sigma \ll \omega$ and these values are not small; they are offered as a sense of scale for the defaults, not as a design guarantee.

The bands are owned by the controller channels: for the parallel PID, $|C(j\omega)| \approx K_i/\omega$ at low frequency, $\approx K_p$ in the mid band, $\approx K_d \omega$ at high frequency. Hence N_0

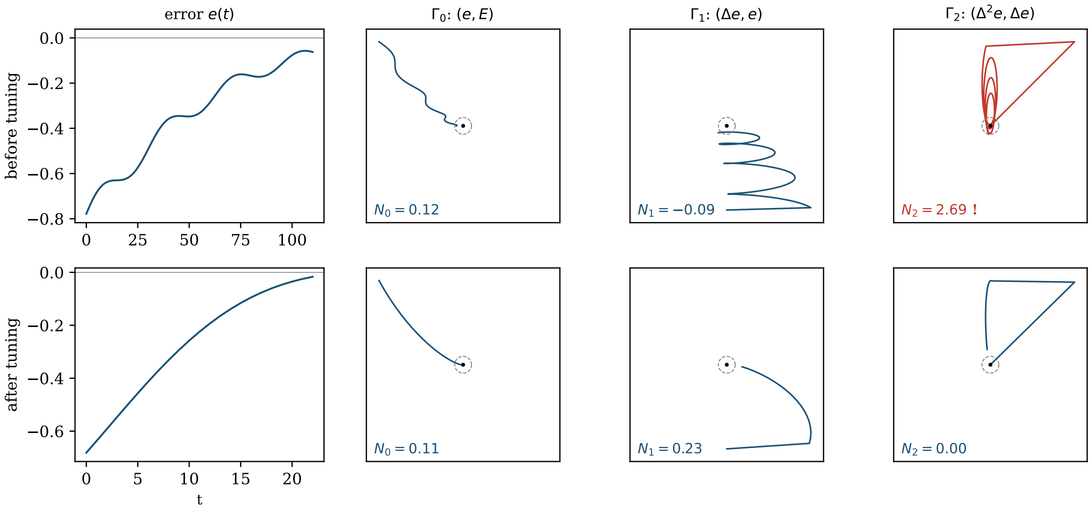

Figure 1: The method at a glance, on plant P2 (Section 5). *Top row (before tuning):* the step-response error looks unremarkable, yet Γ_2 loops 2.69 times—well past its limit of 1.0—revealing a fast mode invisible in $e(t)$ itself. This is row 3 of Table 1: cut D. *Bottom row (after tuning):* all three portraits spiral cleanly into the settling point without a loop.

indicts K_i, N_1 indicts K_p, N_2 indicts K_d. This correspondence is the entire basis of the tuning rule.

Leakage, and its direction. Band selectivity must not be over-read: it governs where a mode is *most* visible, not where it is visible at all. Differencing a damped oscillation yields another damped oscillation with the same σ and ω—the same spiral, hence essentially the same winding count—so a single mode contributes its count to *every* portrait it manages to dominate. And dominance spreads upward: each differencing step multiplies the mode by ω but the monotone settling trend only by its slower decay rate, so an oscillatory mode becomes *more* dominant along the derivative chain, while integration suppresses it. A mid-band mode therefore leaks its full count into Γ_2; a high-band mode leaves Γ_1 and Γ_0 untouched. Synthetically: a settling trend plus one under-damped mid mode alone evaluates to $\mathbf{N} = (0.12, 3.9, 6.5)$—the *high* count exceeds the mid one, purely by leakage—while a lone high mode of comparable severity gives $\mathbf{N} = (0.12, 0.05, 5.7)$, leaking nowhere below. Two consequences follow. Downward cleanliness means the *lowest* violated band is always genuinely guilty: nothing exists below it to leak into it. Upward contamination means violations *above* the lowest are unattributable from the counts alone: a two-mode error and a single strong mid mode produce indistinguishable feature vectors. The tuning rule of Section 4 is built around exactly this asymmetry.

3. Well-posedness of the count

The counts of Definition 1 conceal a subtlety that, to our knowledge, affects every response feature built on differenced signals. Per-axis normalization is what buys Proposition 1; but it also means each portrait is blind to the *absolute* scale of its coordinates. If the error retains a persistent micro-oscillation—at a fraction of a percent of the step, from quantization, a marginally damped parasitic mode, or numerical residue—that ripple can dominate the differenced coordinates Δe, $\Delta^2 e$ long after e itself has settled. Normalized to its own peak, the ripple fills the portrait and winds indefinitely: the count then grows without bound as the observation window lengthens, and comparisons against fixed thresholds become artefacts of the window.

This is not hypothetical. For the plant $P(s) = e^{-4s}/(8s+1)^6$ at gains reached mid-iteration by the tuning rule, the response settles to $|e| < 0.005$ (peak 1.0) well inside the default window, yet the counts evaluate to $\mathbf{N} = (0.12, 1.84, 2.86)$ on that window, $(0.62, 9.22, 17.2)$ on a window six times longer, and $(0.62, 17.3, 37.3)$ on twice that: N_1, N_2 grow essentially linearly in window length. In closed loop the consequence is erratic: threshold comparisons flip under 10% gain changes and the corrective moves of the tuning rule fight each other without converging. Figure 2 shows the pattern: unguarded counts climb without bound as the window lengthens, while the guarded counts defined next sit flat.

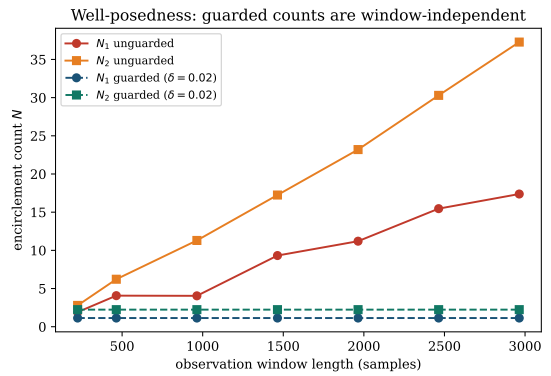

Figure 2: Window dependence of N_1, N_2 on the counterexample plant, guarded (Definition 2) versus unguarded (Definition 1 applied to the raw window). Guarded counts are exactly flat.

The remedy is to anchor the window to the settling of the parent signal.

Definition 2 (Settling-anchored window)**.** *Fix $\delta \in (0,1)$ (default $\delta = 0.02$). Let $k_\delta = 1 + \max\{k : |e_k| > \delta \max_j |e_j|\}$ and truncate all three trajectories* (1) *at k_δ before applying Definition 1.*

Proposition 3 (Well-posedness)**.** *Under Definition 2, the features* $\mathbf{N}$ *are independent of the window length for every window containing the δ-settling time of e, and Proposition 1 continues to hold (δ is dimensionless and defined on the normalized error).*

Proof. All samples beyond k_δ are discarded, and k_δ itself depends only on e up to its δ-settling time; enlarging the window changes neither. Invariance is preserved because k_δ is defined through the ratio $|e_k| / \max_j |e_j|$. □

On the counterexample above, the guarded counts evaluate to $\mathbf{N} = (0.12, 1.14, 2.24)$ on all three windows—identical to three digits—and, as Section 5 shows, the closed-loop tuning iteration on the same plant converges cleanly once the guard is in place. The guard is henceforth part of the feature definition; the method thus carries exactly three dimensionless constants $(\bar{\mathbf{N}}, \varepsilon, \delta)$, none of which was adjusted for any plant in this paper.

Remark 1. *Proposition 3 formalizes, and enforces, the practical requirement that each tuning experiment be observed until the loop settles. The failure mode it excludes—unbounded winding of a noise-dominated differenced portrait—is silent and window-dependent, which is precisely why it deserves a formal guard rather than an operator's discretion.*

Remark 2 (Scope of the guard)**.** *Proposition 3 is stated for a noise-free e. With measurement noise of standard deviation σ_n, k_δ is well defined only if the settling band exceeds the noise floor, $\delta \max_j |e_j| \gg \sigma_n$; otherwise k_δ tracks the last noise excursion rather than true settling, and the divergence of Section 3 can reappear despite the guard. The default $\delta = 0.02$ is therefore adequate only while noise stays below roughly two percent of the step amplitude—a condition to be checked, not assumed, in a given deployment. Two further practical points follow the same logic: enough samples should be retained per transient (some tens to a few hundred) that Δe and $\Delta^2 e$ are not dominated by quantization, and the derivative in* (1) *and in the controller is taken on the error, not the measurement, so that a setpoint step does not itself inject a derivative kick into Γ_2.*

4. The tuning rule

The rule is stated in the coordinates in which it was designed. Order the gains by the frequency band each channel dominates and write

$$K^{(0)} = K_i, \qquad K^{(1)} = K_p, \qquad K^{(2)} = K_d, \qquad (2)$$

so that the band index aligns with the portraits: N_k is the feature that indicts $K^{(k)}$. Parametrize the gains as multipliers $F^{(k)} \in [F_{\min}, F_{\max}]$ (defaults $[0.01, 5]$) on any conservative stabilizing initialization, fix a step β (default 0.1), and let $\gamma = 1/(1-\beta)$. Each iteration performs one step test, evaluates $\mathbf{N}$, and updates the gains.

The rule, in words: *scan the portraits from slow to fast; the first one that loops too much names the gain to turn down one notch, and every gain below it turns up one notch; if none loops, turn all three up.* Table 1 is the entire decision logic. The classical commissioning craft is recognizable row by row—slow cycling: back off the reset; ringing: back off the gain; buzzing: back off the rate—with the compensating increases added, and the lower-triangular stencil that gives the rule its name is directly visible in the arrow pattern.

Table 1: The tuning rule at a glance. The first matching row applies, top to bottom, once per step test; "loops" means $N_k > \bar{N}_k$. ↑: multiply by γ; ↓: divide by γ; ·: unchanged. Projections onto $[F_{\min}, F_{\max}]$ implied.

First loop	F_i	F_p	F_d	craft reading
Γ_0	↓	·	·	slow cycling: less reset
Γ_1	↑	↓	·	ringing: less gain
Γ_2	↑	↑	↓	buzzing: less rate
none	↑	↑	↑	all quiet: tighten

Procedure.

1. Close the loop with any conservative, stabilizing gains.
2. Step the setpoint; record the error e until it settles.
3. Truncate the record where $|e|$ last leaves 2% of its peak (Definition 2).
4. Form the three portraits (Fig. 1) and count loops.
5. Apply the matching row of Table 1.
6. Repeat; once the gains begin to alternate, keep the last tuning that violated no loop limit.

Table 2: Constants and defaults, used unchanged on every plant in this paper.

Symbol	Meaning	Default
$\bar{\mathbf{N}}$	loop limits	$(0.5, 0.75, 1.0)$
ε	truncation radius	0.1
δ	settling band	0.02
β	step size	0.1
box	multiplier range	$[0.01, 5]$

Formally, with the *violated set* and its lowest member

$$V = \{k : N_k > \bar{N}_k\}, \qquad k_{\min} = \min V \tag{3}$$

($k_{\min} = 3$ by convention if $V = \emptyset$), the update reads

$$F^{(k)} \leftarrow \begin{cases} \gamma F^{(k)}, & k < k_{\min} \text{ (raise band)}, \\ F^{(k)}/\gamma, & k = k_{\min} \text{ (cut band)}, \\ F^{(k)}, & k > k_{\min} \text{ (untouched)}, \end{cases} \tag{4}$$

for $k_{\min} = 3$ degenerating to raising all bands. Only the lowest violated band is acted on, even when $|V| > 1$: by the leakage asymmetry of Section 2, $k_{\min}$ is the only member of V whose guilt is unambiguous, while the counts above it may be leakage, genuine, or a mixture, in proportions the features cannot resolve. Bands above $k_{\min}$ are therefore left alone—not out of caution, but because there is no evidence on which to act.

The rule admits a one-sentence description: the loop-gain profile is pushed up from the low-frequency end like a rising level, halting and stepping down at the first band that objects. Equivalently, the iteration is a *lexicographic bandwidth maximizer*: it raises loop gain in all bands subject to the damping constraints $N_k \leq \bar{N}_k$ ordered by band, always yielding to the lowest-band complaint first. It is thus a fixed-point search for the boundary of the well-damped set $\mathcal{F} = \{\theta : \mathbf{N}(\theta) \leq \bar{\mathbf{N}}\}$.

Three structural facts follow directly from (4). First, the priority order and the direction of gain redistribution are the same fact seen twice: because violations are checked from the lowest band up, every band into which the rule pours gain ($k < k_{\min}$) was verified clean *in the current experiment*—the triangular shape is what licenses the compensating increases. Second, in logarithmic gain coordinates the three corrective moves ($k_{\min} = 0, 1, 2$) form a lower triangular matrix with unit diagonal magnitude—unimodular—so their compositions generate every gain combination on the multiplicative γ-lattice: the moves are individually crude but jointly complete, and a compensation that proves unwanted is cancelled exactly by a lower-band correction one iteration later, at the cost of one experiment. Third, the iteration is an *implicit attribution test*: after the cut at $k_{\min}$, whatever violations clear together with it were leakage of the $k_{\min}$ mode, and whatever persists is genuine and becomes the new $k_{\min}$. Sequential single-band correction is thus not a slower version of correcting all violated bands at once; it is the inference procedure that separates leaked from genuine violations, one real experiment per question. Cutting all of V simultaneously would forgo exactly this inference and penalize channels for violations that are largely not their own; Section 5 shows this occurring on the battery.

Remark 3. *Nothing in* (4) *is specific to three channels. Any controller whose channels dominate disjoint, ascending frequency bands—PI, PID with a second derivative term, a lead-lag chain—admits the identical construction: m band-ordered channels, m portraits obtained by walking one step further along the derivative chain per band, the lowest violated band, and the same triangular stencil.*

Assumption 3 (Band monotonicity)**.** *In the region traversed, each feature is locally non-decreasing in its associated band gain (N_k in $K^{(k)}$, $k = 0, 1, 2$), and the corrective rule* (4)*, applied where band $k_{\min}$ is violated, does not increase $N_{k_{\min}}$.*

Proposition 4 (Behaviour of the iteration)**.** *Unconditionally, the iteration is deterministic and its iterates remain in the multiplier box. Under Assumption 3, from any $\theta^0 \in \mathcal{F}$ the iteration reaches every neighbourhood of $\partial\mathcal{F}$ in finitely many steps—or saturates at the box, if $\mathcal{F}$ contains it—and thereafter remains within a band of $\partial\mathcal{F}$ of width one multiplicative step per gain, on which it executes a limit cycle.*

Proof. Boundedness is by construction of the projections. On $\mathcal{F}$, the rule ($V = \emptyset$) multiplies all gains by

$\gamma > 1$, an expanding map that leaves any compact subset of the interior in finitely many steps or saturates at $F_{\max}$. Once $V \neq \emptyset$, the corrective form of the rule reduces the lowest violated feature by a fixed factor per step (Assumption 3); when it clears, either the leaked violations above it clear with it or a genuine one becomes the new $k_{\min}$ and is reduced in turn, restoring feasibility in finitely many steps; alternation of expansion and correction confines the trajectory to the stated band. □

Stopping rule. Because the expanding form $V = \emptyset$ is active exactly on $\mathcal{F}$, the terminal iterate of a fixed-length run may be infeasible; the algorithm therefore stops on cycle detection or an iteration cap and returns *the last iterate with* $\mathbf{N} \leq \bar{\mathbf{N}}$—always well damped, within one step of the fastest such tuning found. Assumption 3 encodes the classical monotonicity of loop damping in channel gain, and within the FOPTD class of Assumption 1 it holds between a conservative initialization and the stability boundary; outside that class—strongly non-minimum-phase or resonant dynamics—it can fail, in which case the iteration parks at a gain bound: detectable, safe, unproductive. We state the assumption rather than a slogan, and report below that it held on every plant tested. Safety in operation is inherited from the step size: gains move $\beta = 10\%$ per iteration, and a step test aborts with rollback if $|e|$ exceeds a configured multiple of the step size.

Remark 4 (Bumpless implementation)**.** *Because every iteration changes K_p, K_i, K_d while the loop remains closed, the controller should be realized in* velocity *(incremental) form,*

$$\Delta u_k = K_p(e_k - e_{k-1}) + K_i T_s e_k + \tfrac{K_d}{T_s}(e_k - 2e_{k-1} + e_{k-2}),$$
$$u_k = u_{k-1} + \Delta u_k, \tag{5}$$

rather than the positional form of Assumption 2 evaluated directly. The velocity form commands only an increment on top of whatever control value is currently applied, so a change of gains at an iteration boundary—or an initial transfer from manual to automatic control—produces no discontinuity in u: there is no integrator state to reinitialize and no term whose value jumps when K_i or K_d is rescaled. This bumpless property is what allows the gains to be updated between step tests without perturbing the plant through the actuator itself, and is assumed throughout the experiments of Section 5.

5. Validation on a plant battery

The method—guarded features, rule of Table 1, stopping rule—was run on four process-typical plants spanning lag-dominant to delay-dominant and high-order dynamics:

$$\begin{aligned}
\text{P1 (lag-dominant):}\quad & \tfrac{e^{-s}}{(10s+1)(s+1)^3},\\
\text{P2 (balanced):}\quad & \tfrac{1.25\,e^{-8s}}{(5s+1)^4},\\
\text{P3 (delay-dominant):}\quad & \tfrac{e^{-10s}}{(2s+1)(s+1)^3},\\
\text{P4 (high-order slow):}\quad & \tfrac{e^{-4s}}{(8s+1)^6}.
\end{aligned}$$

All four satisfy Assumption 1: their open-loop step responses are monotone with dead time, even where their order exceeds one (Table 3). The runs use the identical constants $\bar{\mathbf{N}} = (0.5, 0.75, 1.0)$, $\varepsilon = 0.1$, $\delta = 0.02$, $\beta = 0.1$, box $[0.01, 5]$, throughout. In each case the simulated plant serves exclusively as a stand-in for the physical process: it generates step responses; the tuner sees only the recorded error, exactly as on hardware. Initial gains were a nominal design with proportional and integral gains halved and derivative gain tripled—a deliberately mis-tuned start exercising the rule at every band position.

Three observations. First, on the plants where the mis-tuned derivative channel produces a fast parasitic mode (P2 and P4), the diagnosis is invisible in the raw step response but explicit in Γ_2 ($N_2 = 2.7$ initial for P2, with the lower bands clean, so the indictment is unambiguous); the rule repeatedly cuts band 2, trading D for P and I, until the portrait clears, and P2 first reaches feasibility at iteration 34. Second, on every plant the tail of the run exhibits the limit cycle of Proposition 4, with feasibility revisited within a few iterations of the end of the run (final column)—including for P4, whose iteration churned indefinitely *without* the well-posedness guard of Section 3 and converged cleanly with it. Third, P1 demonstrates the saturation case: within a $[0.01, 5]$ box this loop admits no encirclement at all, and the method honestly returns the box bound rather than a fictitious boundary.

The battery also exhibits the leakage asymmetry of Section 2, and with it the attribution test of Section 4, live. At iteration 16 of the P2 run the features read $\mathbf{N}$ = (0.11, 2.08, 2.22)—bands 1 and 2 violated with nearly equal counts. A single cut of band 1 changes this to (0.11, 0.19, 1.25): the mid-band violation is gone, and almost half of the high-band count has vanished *with* it—that half was leakage of the mid-band mode—while a genuine high-band remainder of 1.25 persists and becomes the next $k_{\min}$. A rule that had cut both members of V on the strength of the raw counts would have penalized the derivative channel for a violation that was largely not its own. We verified this consequence directly: running the battery with a variant that cuts ev-

Table 3: Plant battery: initial and returned features, returned multipliers, iteration at which the returned (last feasible) tuning occurred, out of 120. Identical constants throughout; rule (4).

Plant	**N** initial	**N** returned	(F_p, F_i, F_d)	iter
P1	$(0.12, 0.13, 0.26)$	$(0.11, 0.17, 0.30)$	$(5.00, 2.87, 5.00)^*$	118
P2	$(0.12, -0.09, 2.69)$	$(0.11, 0.23, 0.00)$	$(1.69, 2.32, 0.25)$	114
P3	$(0.12, 0.09, 0.00)$	$(0.46, 0.68, 0.77)$	$(1.69, 2.87, 0.59)$	112
P4	$(0.12, 0.11, 0.25)$	$(0.12, 0.23, 0.00)$	$(1.37, 2.09, 0.39)$	119

*Multiplier box reached: no encirclement occurs within the box for this fast, lag-dominant loop, and the iteration returns the box bound—the saturation case of Proposition 4.

ery violated band simultaneously drives P2's derivative multiplier to 0.04—the channel is nearly extinguished by leaked evidence—and degrades the returned tuning (IAE 10.7 against 6.8 for rule (4)), with a similar degradation on P3. Simultaneous multi-band correction is thus not adopted: it is faster to feasibility from severely mis-tuned starts, but it acts on counts the features cannot attribute.

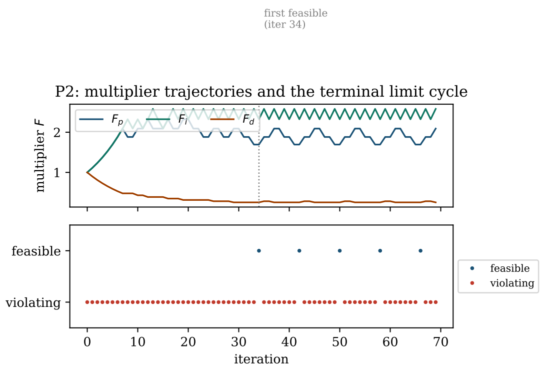

Figure 3: P2, single-band rule: multiplier trajectories (top) and feasibility per iteration (bottom). After an initial corrective phase, the run settles into the boundary limit cycle of Proposition 4, alternating between violating and feasible iterates.

6. Limits

The method presumes band-wise oscillation is curable by cutting the gain that owns the band (Assumption 3); strong non-minimum-phase or resonant dynamics can break this, visibly and safely, by parking the iteration at a gain bound. The rule as stated reacts only to $N_k > \bar{N}_k$; the sign of N_k is otherwise unused, though a negative count (clockwise winding, observed for P2 in Table 3) is itself a signature of inverse step response and so, in principle, evidence bearing on Assumption 1—an observation left for future work rather than built into the present rule. Measurement noise enters the differenced coordinates directly; the guard of Definition 2 excludes settled noise under the condition of Remark 2, not broadband noise within the transient, and deployment on noisy measurements requires filtering deliberately outside this paper's scope. Each iteration requires one settled, undisturbed step-response window; a disturbance corrupts one test, not the procedure. The rule acts on one band per iteration, and this is the price of attribution rather than mere caution: violations above the lowest cannot be told apart from leakage until the lowest is cleared (Section 2), so a severely mis-tuned start that violates all bands converges through a sequence of single corrections—starting P4 at $(2K_p, 3K_i, 4K_d)$, with all three counts near 2.9, the rule needs 74 iterations to first reach feasibility. When every band screams at once the counts typically reflect one dominant mode leaking everywhere, and a uniform reduction of all gains—an attribution-free emergency backoff—is a sound faster response; we keep the rule minimal and leave this to the operator or to future work. Leakage can also misdirect a single correction when the true offender sits just below its own threshold while its leaked count trips a higher band; the stencil self-corrects, since the accompanying raises push the true offender over its own threshold within a few iterations, but at the cost of those iterations. The destination is a fast well-damped tuning, not an optimum; where a criterion matters, this output is a sound starting point for criterion-based refinement, not a substitute for it.

7. Conclusion

Manual PID tuning survives because it asks little and shows its work; we kept both properties and removed the subjectivity. Three portraits of the error make under-damping visible band by band; a winding count makes the visibility a number; a well-posedness guard makes the number window-independent; one triangular rule, cutting only the band whose blame is unambiguous, makes it a decision. The features carry no units and no clock, so three fixed dimensionless constants serve every plant tested. The iteration is deterministic and bounded unconditionally, and converges to the boundary of the well-damped set under one explicit, classical assumption. Future work: principled filtering of the differenced portraits under measurement noise, band-selective acceleration from sub-threshold winding, attribution-aware multi-band correction (separating leaked from genuine violations above the lowest band, e.g., by comparing dominant periods across portraits), and the multi-channel construction of Remark 3 applied to 2-DOF and cascade structures, where the band ordering across layers is less immediate.

Data availability

Reference implementation, simulation code, and scripts reproducing every result: `https://github.com/ottapav/roboPID-simulator`.

References

[1] J.G. Ziegler, N.B. Nichols, Optimum settings for automatic controllers, Trans. ASME 64 (1942) 759–768.

[2] K.J. Åström, T. Hägglund, The future of PID control, Control Eng. Pract. 9 (11) (2001) 1163–1175.

[3] G.H. Cohen, G.A. Coon, Theoretical consideration of retarded control, Trans. ASME 75 (1953) 827–834.

[4] D.E. Rivera, M. Morari, S. Skogestad, Internal model control: PID controller design, Ind. Eng. Chem. Process Des. Dev. 25 (1) (1986) 252–265.

[5] K.J. Åström, T. Hägglund, Automatic tuning of simple regulators with specifications on phase and amplitude margins, Automatica 20 (5) (1984) 645–651.

[6] S. Skogestad, Simple analytic rules for model reduction and PID controller tuning, J. Process Control 13 (4) (2003) 291–309.

[7] H. Hjalmarsson, M. Gevers, S. Gunnarsson, O. Lequin, Iterative feedback tuning: theory and applications, IEEE Control Syst. Mag. 18 (4) (1998) 26–41.

[8] M.C. Campi, A. Lecchini, S.M. Savaresi, Virtual reference feedback tuning, Automatica 38 (8) (2002) 1337–1346.

[9] N.J. Killingsworth, M. Krstic, PID tuning using extremum seeking, IEEE Control Syst. Mag. 26 (1) (2006) 70–79.